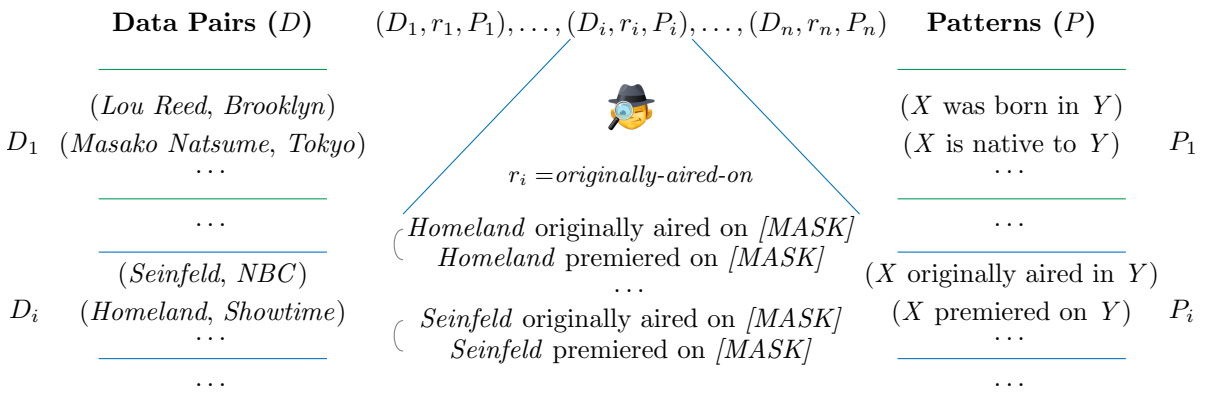

Figure 2: Overview of our framework for assessing model consistency. D_i ("Data Pairs (D)" on the left) is a set of KB triplets of some relation r_i, which are coupled with a set of *quasi-paraphrase* cloze-patterns P_i ("Patterns (P)" on the right) that describe that relation. We then populate the subjects from D_i as well as a mask token into all patterns P_i (shown in the middle) and expect a model to predict the same object across all pattern pairs.

PLM, especially in the context of KBs. However, poor consistency in this setup strongly implies that consistency would be even lower without restricting candidates.

4 The PARAREL🤘 Resource

We now describe PARAREL🤘, a resource designed for our framework (cf. Section 3.2). PARAREL🤘 is curated by experts, with a high level of agreement. It contains patterns for 38 relations[4] from T-REx (Elsahar et al., 2018) — a large dataset containing KB triples aligned with Wikipedia abstracts — with an average of 8.63 patterns per relation. Table 1 gives statistics. We further analyse the paraphrases used in this resource, partly based on the types defined in Bhagat and Hovy (2013), and report this analysis in Appendix B.

Construction Method PARAREL🤘 was constructed in four steps. (1) We began with the patterns provided by LAMA (Petroni et al., 2019) (one pattern per relation, referred to as *base-pattern*). (2) We augmented each base-pattern with other patterns that are paraphrases from LPAQA (Jiang et al., 2020). However, since LPAQA was created automatically (either by back-translation or by extracting patterns from sentences that contain both subject and object), some LPAQA patterns are not correct paraphrases. We therefore only include the subset of correct paraphrases. (3) Using SPIKE (Shlain et al., 2020),[5] a search engine over Wikipedia sentences that supports syntax-based queries, we searched for additional patterns that appeared in Wikipedia and added them to PARAREL🤘. Specifically, we searched for Wikipedia sentences containing a subject-object tuple from T-REx and then manually extracted patterns from the sentences. (4) Lastly, we added additional paraphrases of the base-pattern using the annotators' linguistic expertise. Two additional experts went over all the patterns and corrected them, while engaging in a discussion until reaching agreement, discarding patterns they could not agree on.

Human Agreement To assess the quality of PARAREL🤘, we run a human annotation study. For each relation, we sample up to five paraphrases, comparing each of the new patterns to the base-pattern from LAMA. That is, if relation r_i contains the following patterns: p_1, p_2, p_3, p_4, and p_1 is the base-pattern, then we compare the following pairs $(p_1, p_2), (p_1, p_3), (p_1, p_4)$.

We populate the patterns with random subjects and objects pairs from T-REx (Elsahar et al., 2018) and ask annotators if these sentences are paraphrases. We also sample patterns from different relations to provide examples that are not paraphrases of each other, as a control. Each task contains five patterns that are thought to be paraphrases and two that are not.[6] Overall, we collect annotations for

[4] using the 41 relations from LAMA (Petroni et al., 2019), leaving out three relations that are poorly defined, or consist of mixed and heterogeneous entities.

[5] https://spike.apps.allenai.org/

[6] The controls contain the same subjects and objects so that only the pattern (not its arguments) can be used to solve the task.

# Relations	38
# Patterns	328
Min # patterns per rel.	2
Max # patterns per rel.	20
Avg # patterns per rel.	8.63
Avg syntax	4.74
Avg lexical	6.03

Table 1: Statistics of PARAREL🤘. Last two rows: average number of unique syntactic/lexical variations of patterns for a relation.

156 paraphrase candidates and 61 controls.

We asked NLP graduate students to annotate the pairs and collected one answer per pair.[7] The agreement scores for the paraphrases and the controls are 95.5% and 98.3%, which is high and indicates PARAREL🤘's high quality. We also inspected the disagreements and fixed many additional problems to further improve quality.

5 Experimental Setup

5.1 Models & Data

We experiment with four PLMs: BERT, BERT whole-word-masking[8] (Devlin et al., 2019), RoBERTa (Liu et al., 2019) and ALBERT (Lan et al., 2019). For BERT, RoBERTa and ALBERT, we use a base and a large version.[9] We also report a majority baseline that always predicts the most common object for a relation. By construction, this baseline is perfectly consistent.

We use knowledge graph data from T-REx (Elsahar et al., 2018).[10] To make the results comparable across models, we remove objects that are not represented as a single token in all models' vocabularies; 26,813 tuples remain.[11] We further split the data into N-M relations for which we report determinism results (seven relations) and N-1 relations for which we report consistency (31 relations).

[7] We asked the annotators to re-annotate any mismatch with our initial label, to allow them to fix random mistakes.

[8] BERT whole-word-masking is BERT's version where words that are tokenized into multiple tokens are masked together.

[9] For ALBERT we use the smallest and largest versions.

[10] We discard three poorly defined relations from T-REx.

[11] In a few cases, we filter entities from certain relations that contain multiple fine-grained relations to make our patterns compatible with the data. For instance, most of the instances for the *genre* relation describes music genres, thus we remove some of the tuples were the objects include non-music genres such as 'satire', 'sitcom' and 'thriller'.

Model	Succ-Patt	Succ-Objs	Unk-Const	Know-Const
majority	97.3+-7.3	23.2+-21.0	100.0+-0.0	100.0+-0.0
BERT-base	**100.0**+-0.0	63.0+-19.9	46.5+-21.7	63.8+-24.5
BERT-large	**100.0**+-0.0	**65.7**+-22.1	48.1+-20.2	65.2+-23.8
BERT-large-wwm	**100.0**+-0.0	64.9+-20.3	**49.5**+-20.1	**65.3**+-25.1
RoBERTa-base	**100.0**+-0.0	56.2+-22.7	43.9+-15.8	56.3+-19.0
RoBERTa-large	**100.0**+-0.0	60.1+-22.3	46.8+-18.0	60.5+-21.1
ALBERT-base	**100.0**+-0.0	45.8+-23.7	41.4+-17.3	56.3+-22.0
ALBERT-xxlarge	**100.0**+-0.0	58.8+-23.8	40.5+-16.4	57.5+-23.8

Table 2: Extractability Measures in the different models we inspect. Best model for each measure highlighted in bold.

5.2 Evaluation

Our consistency measure for a relation r_i (*Consistency*) is the percentage of consistent predictions of all the pattern pairs $p^i_k, p^i_l \in P_i$ of that relation, for all its KB tuples $d^i_j \in D_i$. Thus, for each KB tuple from a relation r_i that contains n patterns, we consider predictions for $n(n-1)/2$ pairs.

We also report *Accuracy*, that is, the acc@1 of a model in predicting the correct object, using the original patterns from Petroni et al. (2019). In contrast to Petroni et al. (2019), we define it as the accuracy of the top-ranked object from the candidate set of each relation. Finally, we report *Consistent-Acc*, a new measure that evaluates individual objects as correct only if *all* patterns of the corresponding relation predict the object correctly. *Consistent-Acc* is much stricter and combines the requirements of both consistency (*Consistency*) and factual correctness (*Accuracy*). We report the average over relations, i.e., macro average, but notice that the micro average produces similar results.

6 Experiments and Results

6.1 Knowledge Extraction through Different Patterns

We begin by assessing our patterns as well as the degree to which they extract the correct entities. These results are summarized in Table 2.

First, we report *Succ-Patt*, the percentage of patterns that successfully predicted the right object at least once. A high score suggests that the patterns are of high quality and enable the models to extract the correct answers. All PLMs achieve a perfect score. Next, we report *Succ-Objs*, the percentage of entities that were predicted correctly by at least one of the patterns. *Succ-Objs* quantifies the degree to which the models "have" the required knowledge. We observe that some tuples

Model	Accuracy	Consistency	Consistent-Acc
majority	23.1+-21.0	100.0+-0.0	23.1+-21.0
BERT-base	45.8+-25.6	58.5+-24.2	27.0+-23.8
BERT-large	48.1+-26.1	**61.1**+-23.0	**29.5**+-26.6
BERT-large-wwm	**48.7**+-25.0	60.9+-24.2	29.3+-26.9
RoBERTa-base	39.0+-22.8	52.1+-17.8	16.4+-16.4
RoBERTa-large	43.2+-24.7	56.3+-20.4	22.5+-21.1
ALBERT-base	29.8+-22.8	49.8+-20.1	16.7+-20.3
ALBERT-xxlarge	41.7+-24.9	52.1+-22.4	23.8+-24.8

Table 3: Knowledge and Consistency Results. Best model for each measure in bold.

are not predicted correctly by any of our patterns: the scores vary between 45.8% for ALBERT-base and 65.7% for BERT-large. With an average number of 8.63 patterns per relation, there are multiple ways to extract the knowledge, we thus interpret these results as evidence that a large part of T-REx knowledge is not stored in these models.

Finally, we measure *Unk-Const*, a consistency measure for the subset of tuples for which no pattern predicted the correct answer; and *Know-Const*, consistency for the subset where at least one of the patterns for a specific relation predicted the correct answer. This split into subsets is based on *Succ-Objs*. Overall, the results indicate that when the factual knowledge is successfully extracted, the model is also more consistent. For instance, for BERT-large, *Know-Const* is 65.2% and *Unk-Const* is 48.1%.

6.2 Consistency & Knowledge

In this section, we report the overall knowledge measure that was used in Petroni et al. (2019) (*Accuracy*), the consistency measure (*Consistency*), and *Consistent- Acc*, which combines knowledge and consistency (*Consistent-Acc*). The results are summarized in Table 3.

We begin with the *Accuracy* results. The results range between 29.8% (ALBERT-base) and 48.7% (BERT-large whole-word-masking). Notice that our numbers differ from Petroni et al. (2019) as we use a candidate set (§3) and only consider KB triples whose object is a single token in all the PLMs we consider (§5.1).

Next, we report *Consistency* (§5.2). The BERT models achieve the highest scores. There is a consistent improvement from base to large versions of each model. In contrast to previous work that observed quantitative and qualitative improvements of RoBERTa-based models over BERT (Liu et al., 2019; Talmor et al., 2020), in terms of consistency, BERT is more consistent than RoBERTa and ALBERT. Still, the overall results are low (61.1% for the best model), even more remarkably so because the restricted candidate set makes the task easier. We note that the results are highly variant between models (performance on *original-language* varies between 52% and 90%), and relations (BERT-large performance is 92% on *capital-of* and 44% on *owned-by*).

Finally, we report *Consistent-Acc*: the results are much lower than for *Accuracy*, as expected, but follow similar trends: RoBERTa-base performs worse (16.4%) and BERT-large best (29.5%).

Interestingly, we find strong correlations between Accuracy and Consistency, ranging from 67.3% for RoBERTa-base to 82.1% for BERT-large (all with small p-values $\ll 0.01$).

A striking result of the model comparison is the clear superiority of BERT, both in knowledge accuracy (which was also observed by Shin et al. (2020)) and knowledge consistency. We hypothesize this result is caused by the different sources of training data: although Wikipedia is part of the training data for all models we consider, for BERT it is the main data source, but for RoBERTa and ALBERT it is only a small portion. Thus, when using additional data, some of the facts may be forgotten, or contradicted in the other corpora; this can diminish knowledge and compromise consistency behavior. Thus, since Wikipedia is likely the largest unified source of factual knowledge that exists in unstructured data, giving it prominence in pretraining makes it more likely that the model will incorporate Wikipedia's factual knowledge well. These results may have a broader impact on models to come: Training bigger models with more data (such as GPT-3 (Brown et al., 2020)) is not always beneficial.

Determinism We also measure determinism for N-M relations, i.e., we use the same measure as *Consistency*, but since difference predictions may be factually correct, these do not necessarily convey consistency violations, but indicate non-determinism. For brevity, we do not present all results, but the trend is similar to the consistency result (although not comparable, as the relations are different): 52.9% and 44.6% for BERT-large and RoBERTa-base, respectively.